Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To eliminate the Minor Stoppages At Wheel Alignment Machine at Scorpio TCF."

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**07.05.19**	**23.05.19**
Completion Date	**15.10.19**	**27.10.19**

Current Status: **CLOSED**

Cross Functional Team

No	Description	Name	Designation	Dept.	Phone
1	**Team Leader**	Prateek Srivastava	L-8 O	STCF	7275502900
2	**Mentor**	Tushar Patil	L-5 M	STCF	9922409136
3	**Member 1**	Sagar Kekade	L-8 O	STCF	8237028924
4	**Member 2**	Savkar Mahesh	L-8 O	STCF	8888897833
5	**Member 3**	Jondhale Dattatray	L-7 O	STCF	7709341353
6	**Member 4**	Satish Patil	L-7 O	STCF	9922448571
6	**Member 5**	Satish Pagar	L-8 O	STCF	9822209653

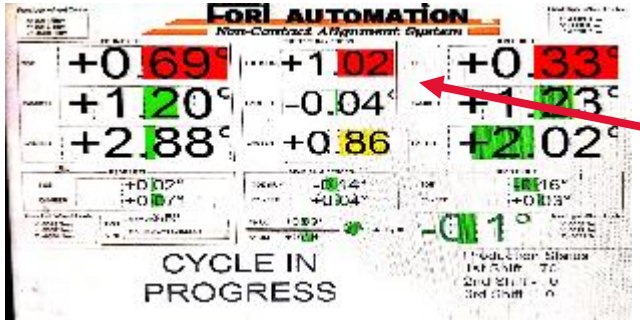
Step1. Select Subject

PROJECT TITLE :-

To reduce the Wheel Alignment machine minor stoppages issues by 60% at CAI

Brief Description :

- **What** : Wheel Alignment Machine not working properly
- **Where** : At CAI , before Roll & Brake Test Stage
- **When** : Since Feb-19
- **Who** : by CAI Associate & officers
- **Why** : Major irritant on line, STR loss.
- **How** : 2 avg. incidences per day of Wheel Alignment system not working which leads to line loss.



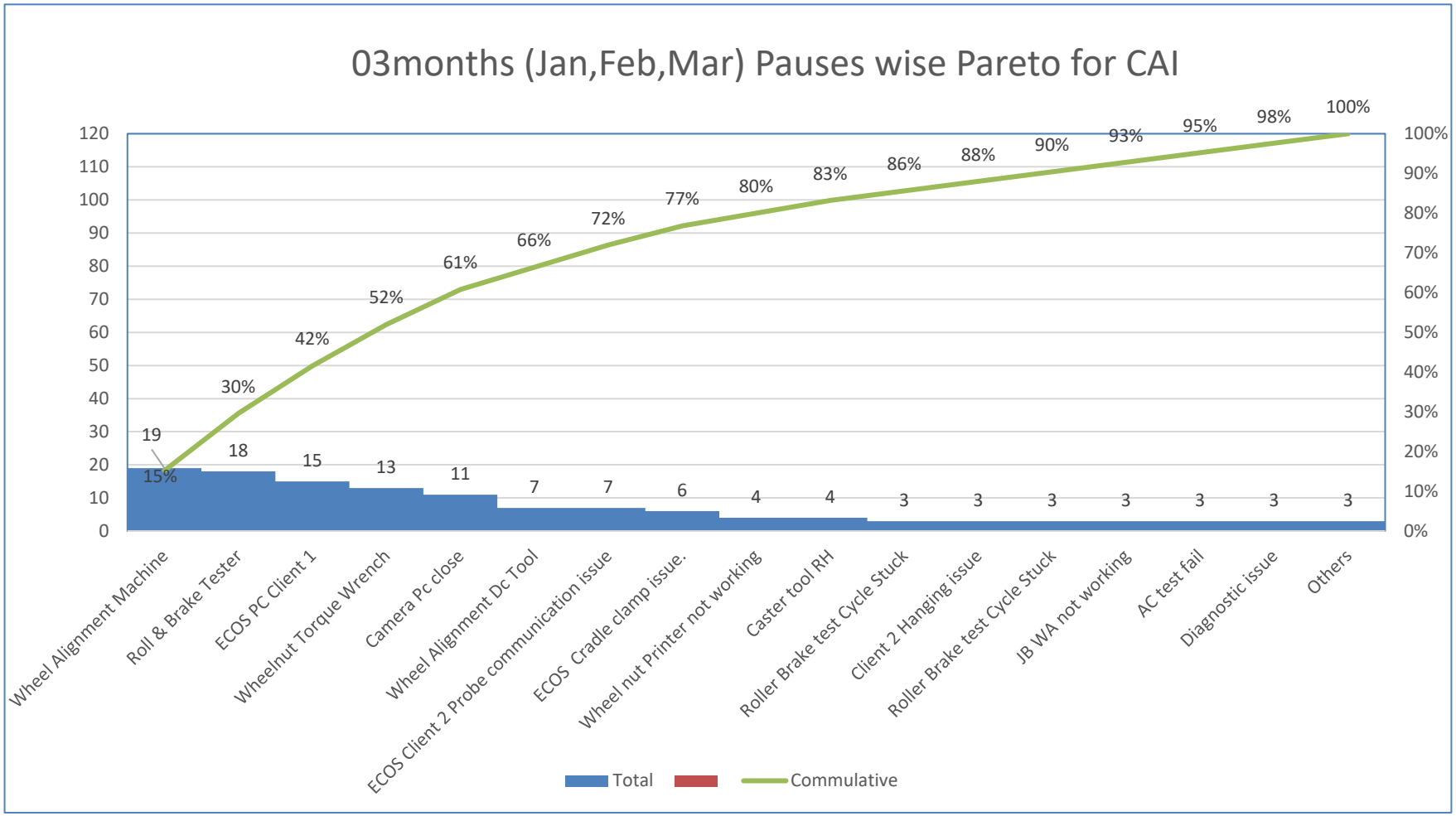
Wheel Alignment Machine not working properly

2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice



Incidences →

Concern →

<u>INFERENCE :-</u>AS PER PARETO GRAPH Wheel Alignment ISSUE IS TOP in minor stoppages list of CAI Area

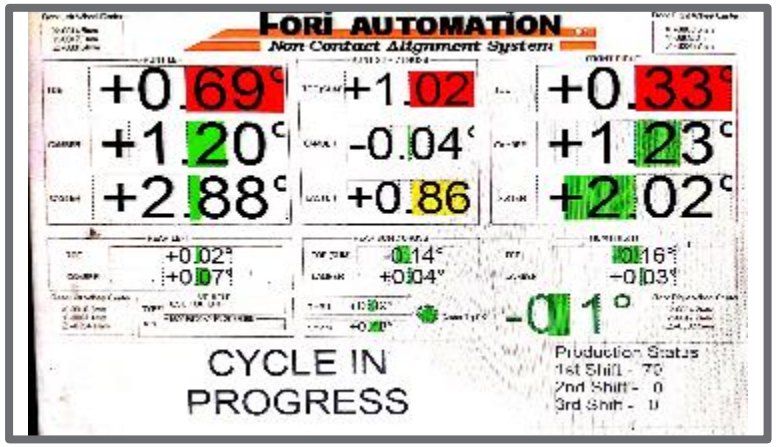
2. Justify the choice

COPQ Cost Calculation -

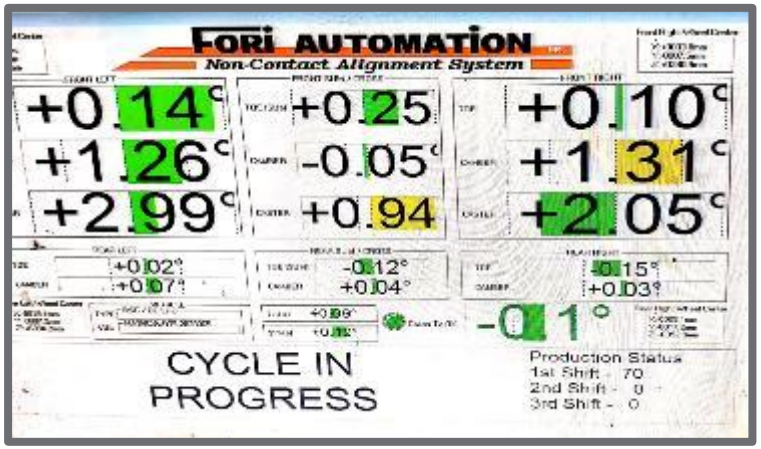
Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	2.0
Total Required Rework time per day	Min	20
Effective working time	Min	465
Operator required /day	nos	0.04
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	84.90
Avg no. of working days	days	25
Operator cost /month	Rs	2122.6
Operator Cost/year	Rs	25471.0

2.b Immediate Containment Action

- ❑ Network Cable plugged out and then again inserted to run wheel Alignment Machine
- ❑ Machine Software restarted to start system



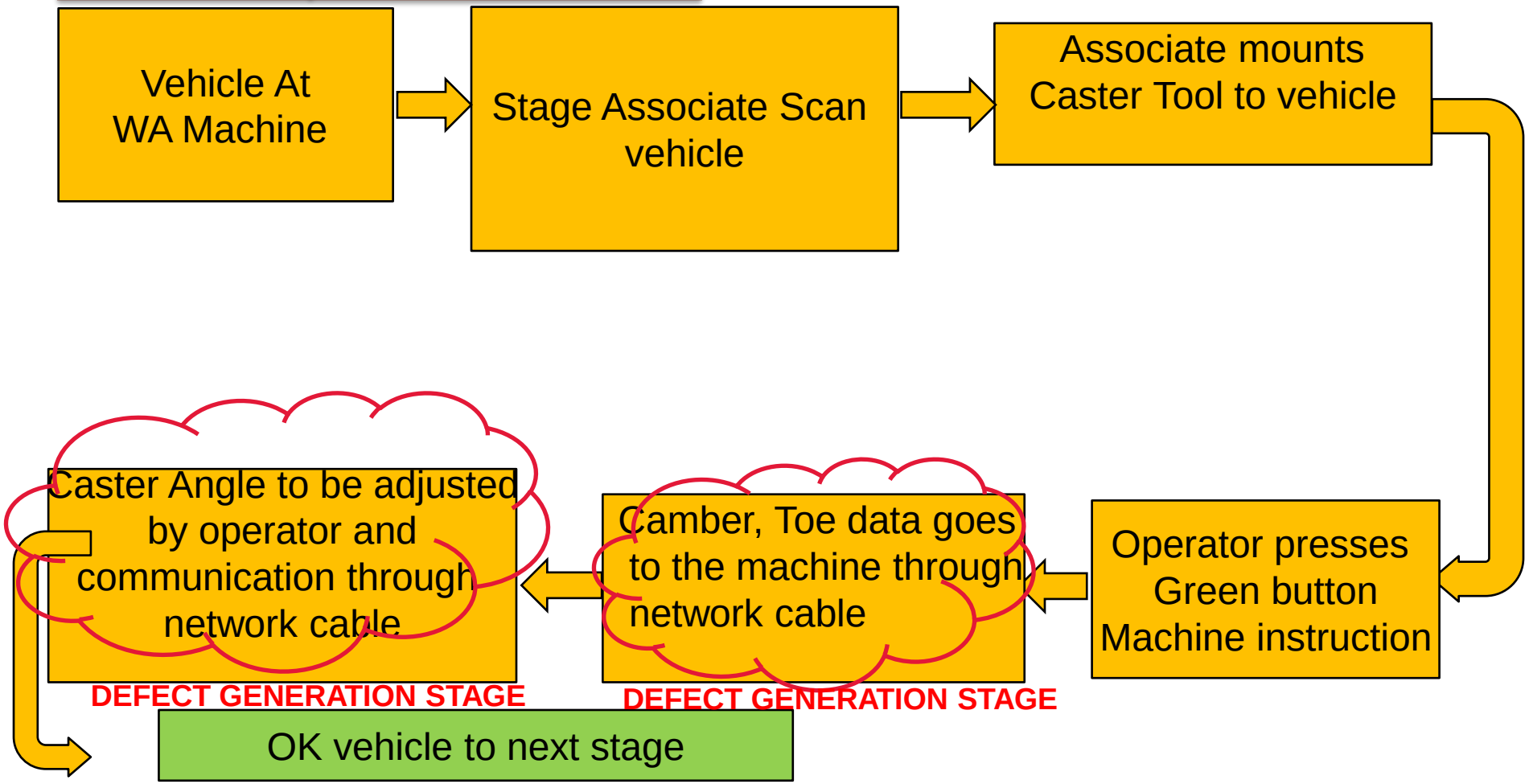
System Stuck



System Started

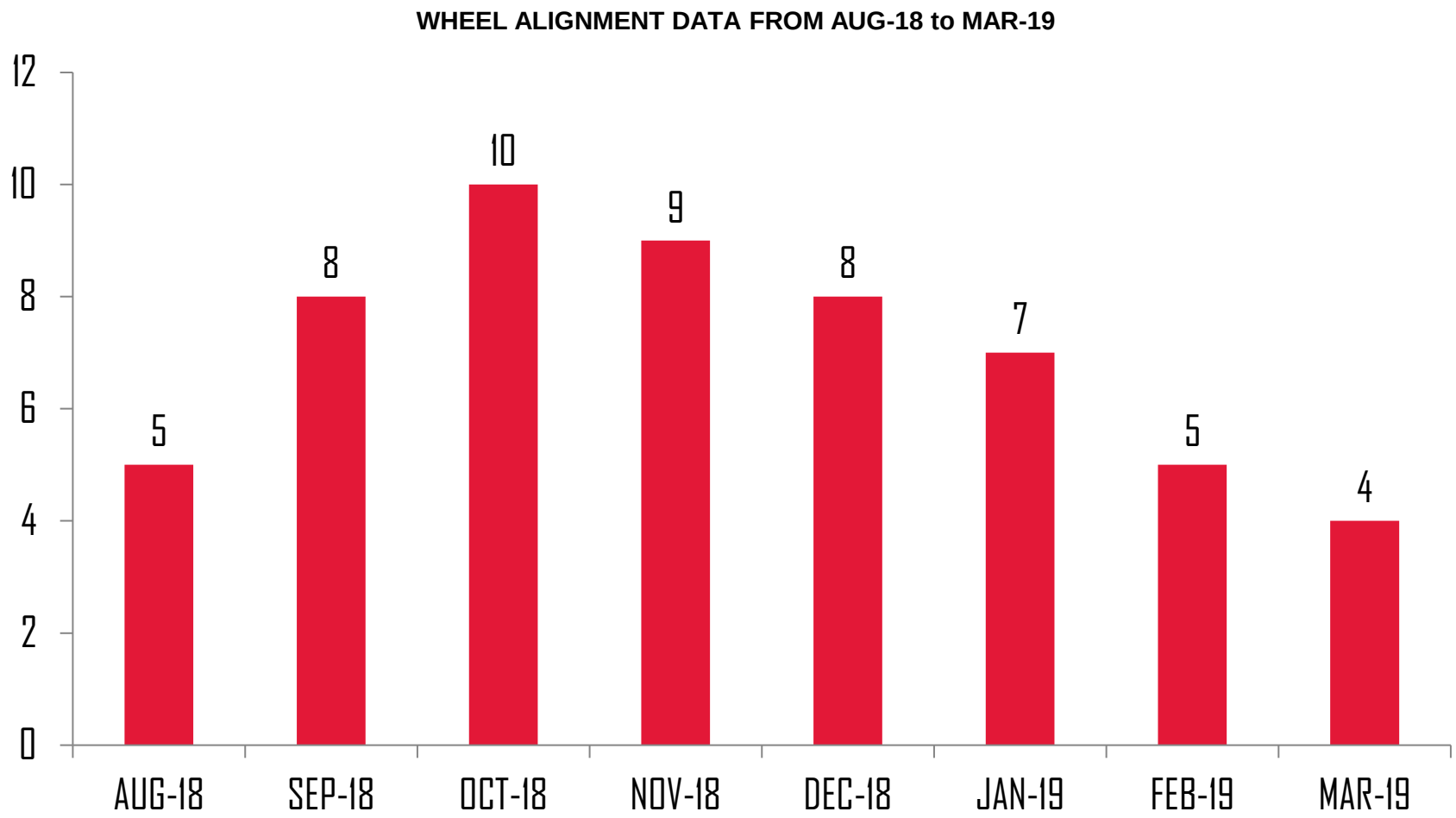
Step 3. Understanding Current Situation

Observation of process:-



3. Understand the current situation

Month wise Trend



INFERENCE :- AS PER MONTH ON MONTH DATA COLLECTED THE SAID DEFECT IS CONTRIBUTING AVERAGE 7 concerns/month

Step 3. Understanding Current Situation

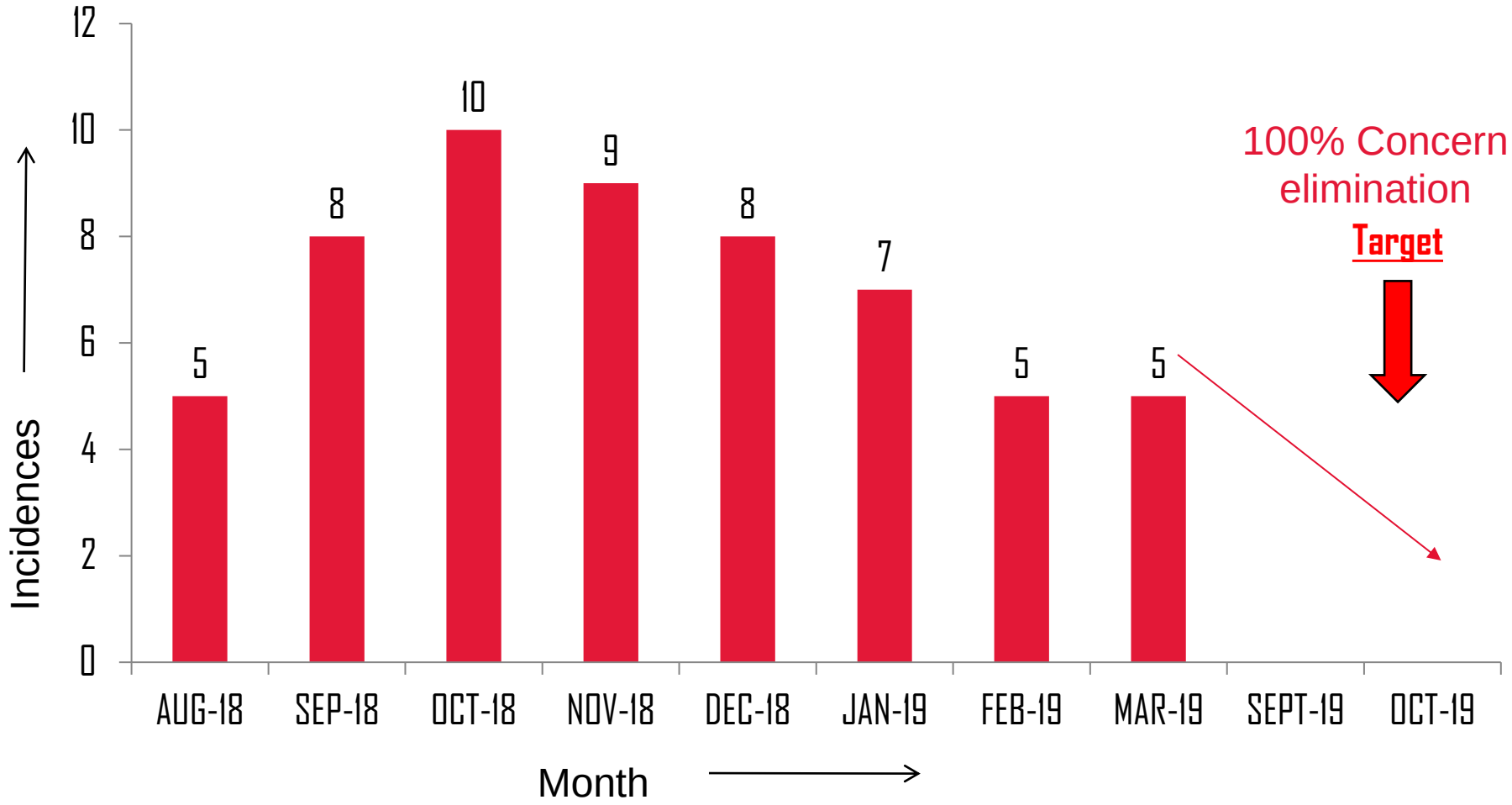
Good & Not So good observations

- Issue Resolve within 8 to 14 min.
- Issue reported in both the shifts
- Issue occurs frequently
- Reasons for stoppage is similar

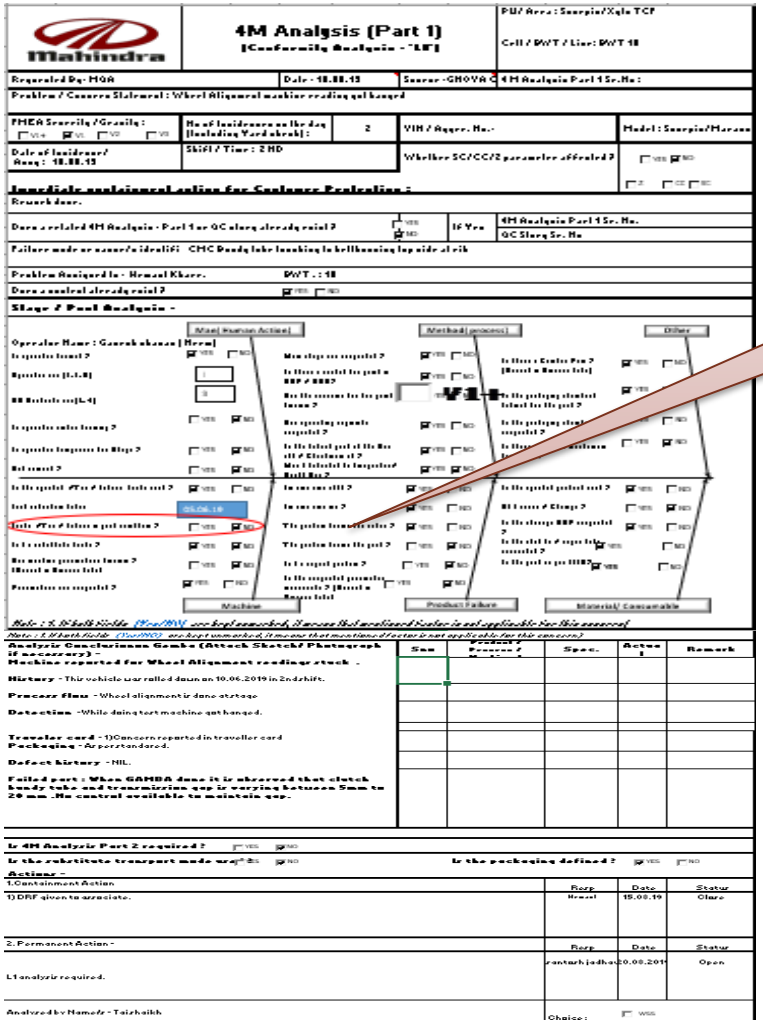
4. Select targets

Present level :- cases reported in last 10 months

Target :- Concern elimination by END OF October-2019



4M Analysis



Inference from L0,
Problem follows with machine,
need detail analysis.

4M analysis Good and Not so good observations	
MAN	Operator Dext. Level L3
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Problem follows to M/C
Product failure	Problem follows vehicle
Material	Handling and shelf life of part adhered

NO QUICK WINS

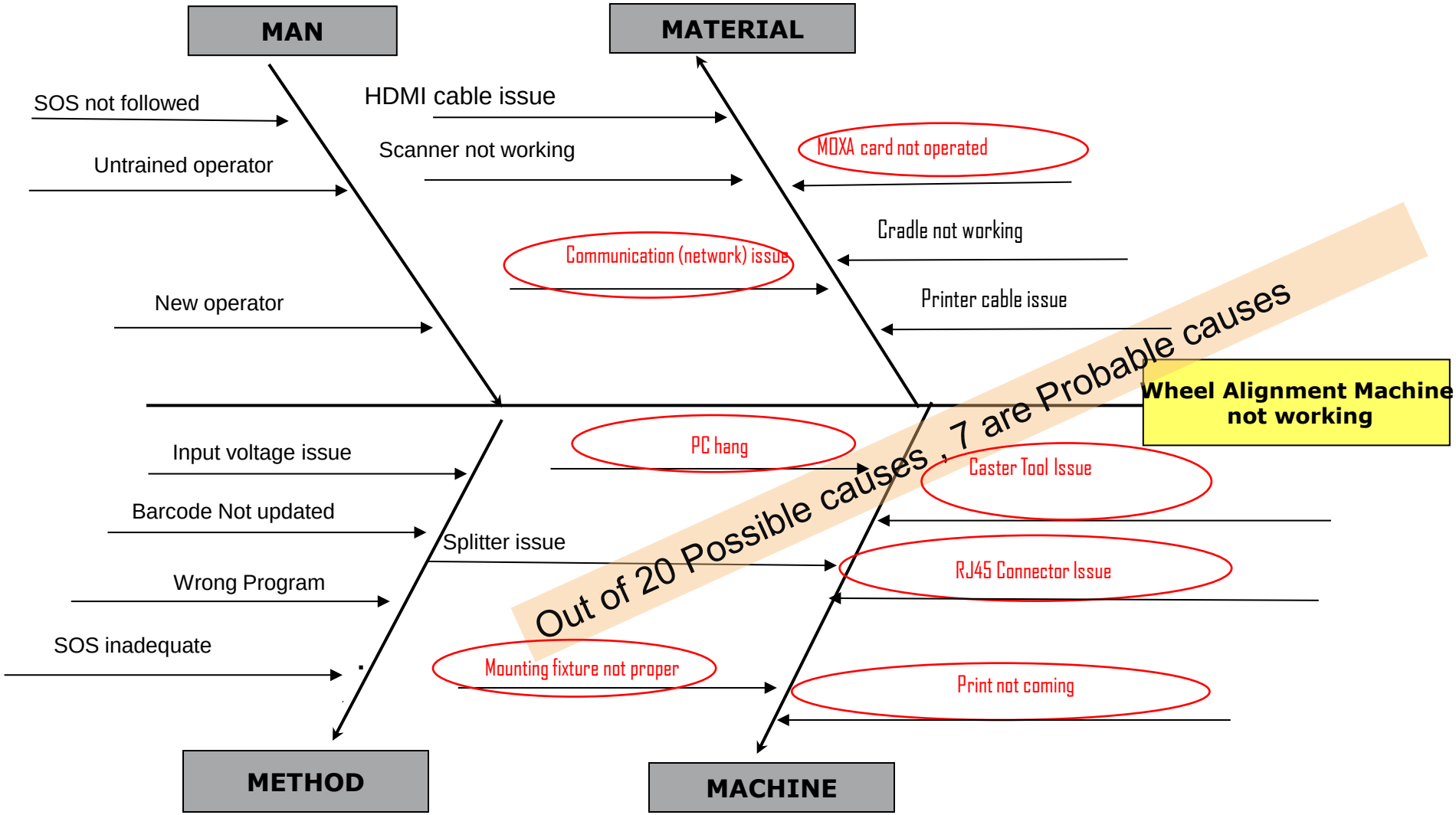
Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr no	Possible cause	Sr. no	Possible cause
1	New Operator	11	OBD pin Broken
2	Untrained operator	12	Communication Issue
3	SOS not followed	13	RJ45 damage
4	SOS inadequate	14	Display Issue
5	Mounting Not Proper	15	Print not coming
6	Scanner not working	16	PC hang
7	Cradle not working	17	MOXA card not operated
8	High Readings	18	Barcode not updated
9	Input voltage issue	19	Printer cable issue
10	Caster Tool Issue	20	Splitter Issue

Step 5. Analysis

List Of Possible Causes



Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr no	Possible Causes	Prateek	Tushar	Sagar	Mahesh	Nilesh	Dattatray	Satish	Total
1	New Operator	1	0	0	0	3	3	3	10
2	Untrained operator	3	3	0	0	3	3	3	15
3	SOS not followed	3	3	0	0	3	1	0	10
4	SOS inadequate	9	0	0	1	3	0	0	13
5	Pipe not proper	3	3	0	0	3	3	3	15
6	Scanner not working	3	0	0	0	3	0	3	9
7	Cradle not working	0	0	3	0	3	3	0	9
8	Communication (Network) Issue	9	9	3	9	3	9	3	45
9	Input voltage issue	3	3	3	3	1	0	0	13
10	Caster Tool Wiring issue	3	9	3	9	9	3	9	45
11	Mounting fixture not proper	9	3	9	9	3	3	3	39
12	MOXA card not operated	9	3	3	9	3	9	9	45
13	Print not coming	9	3	9	9	3	3	3	39
14	Display issue	3	1	3	0	0	0	3	10
15	RJ45 connector issue	9	9	3	3	9	9	3	45
16	PC hang	9	9	9	9	3	3	9	51
17	Program Wrong	3	0	3	1	3	3	3	16
18	Barcode not updated	3	1	3	1	3	1	3	15
19	Printer cable issue	1	0	3	3	0	3	1	11
20	HDMI cable issue	0	0	3	3	0	0	0	6

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 39.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr. no	Not considered Causes	Reasons
1	New Operator	NO New operator deployed on this stage
2	Untrained operator	All trained operator
3	SOS not followed	SOS followed
4	SOS inadequate	SOS was OK
5	Mounting Not proper	In audit found ok
6	Scanner not working	Scanner found ok no issue
7	Cradle not working	Cradle found ok no issue
8	Input voltage issue	Voltage as per Specification
9	Display issue	Found ok
10	Program Wrong	Program OK
11	Barcode Not updated	All Barcode updated
12	Printer Cable issue	Cable continuity ok
13	HDMI Cable issue	Cable continuity ok

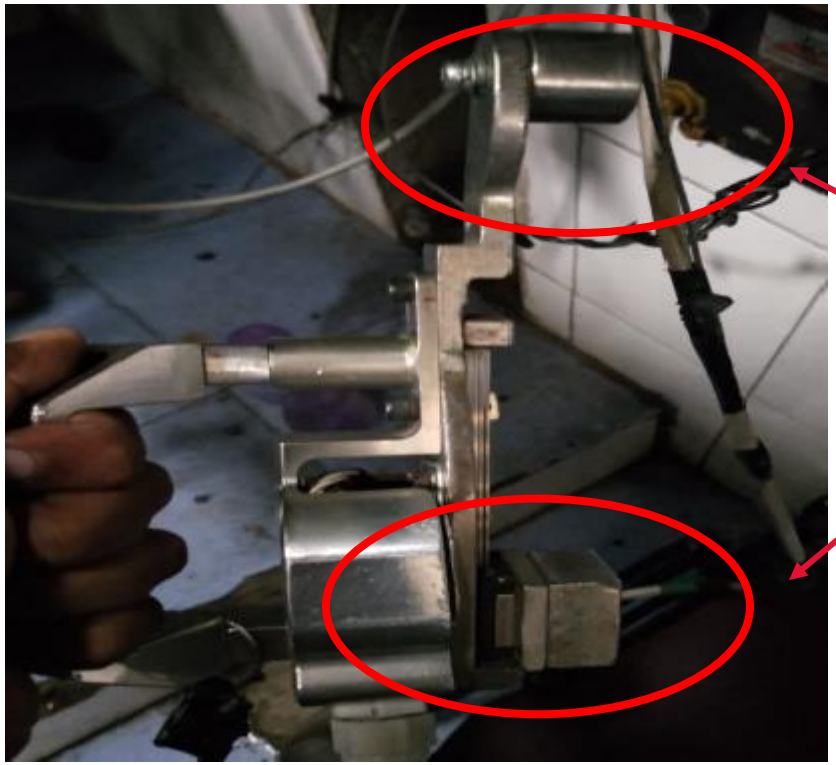
5.1 Identify Probable cause

No	List of Probable Causes
1	Connectivity (Network) Issue
2	Caster Tool wiring Issue
3	Mounting fixture not proper
4	RJ45 connector Issue
5	Print Not coming
6	PC hang
7	MOXA Card not operated

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Mounting fixture not proper	Checked physically for any damage in fixture	No any physical damage found	Hypothesis invalid



Both The fixtures found OK, no physical damage is found

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	RJ45 connector Issue	Visual Check of RJ45 connector and confirm whether it Properly inserted or not.	Connector checked found – Properly inserted Specifications = it should inserted properly	Hypothesis invalid

Connector connected properly

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	MOXA connection	Checking visually for all connectors connected properly or not	Connector checked found – All connectors connected properly Specifications – All LED's should blink	**Hypothesis invalid**

Glowing LED's indicate proper working

TEST OF HYPOTHESIS

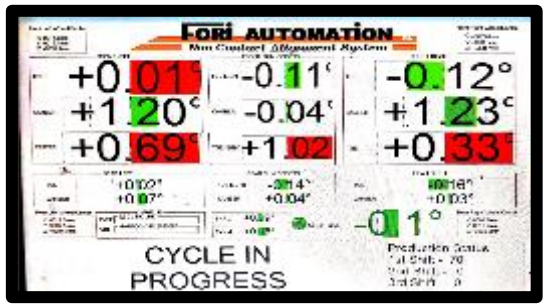
5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Caster Tool Issue	Visual checking with OK ,NOT OK	Observation noted for every month 39% issues due to caster tool	**Hypothesis valid**

Wheel Alignment Machine

Problem Monitoring Chart

Equipment name	Problem/Concern	18-Apr	18-May	18-Jun	18-Jul	18-Aug	18-Sep	18-Oct	18-Nov	18-Dec	18-Jan	18-Feb	18-Mar	18-Apr
Wheel Alignment Machine	New Operator	0	0	0	0	0	0	0	0	0	0	0	0	0
	Untrained Operator	0	0	0	0	0	0	0	0	0	0	0	0	0
	SOS not followed	0	0	0	0	0	0	0	0	0	0	0	0	0
	SOS inadequate	0	0	0	0	0	0	0	0	0	0	0	0	0
	Cradle not working	0	0	0	0	0	0	0	0	0	0	0	0	0
	Input voltage Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Display Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Program Wrong	0	0	0	0	0	0	0	0	0	0	0	0	0
	Barcode not updated	0	0	0	0	0	0	0	0	0	0	0	0	0
	Printer Cable Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDMI Cable Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Connectivity(Network) Issue	2	3	4	2	3	5	4	2	1	3	4	5	2
	Mounting fixture not	0	0	0	0	0	0	0	0	0	0	0	0	0
	Scanner not working	0	0	0	0	0	0	0	0	0	0	0	0	0
	Caster Tool Wiring Issue	1	2	1	2	1	1	1	2	2	1	1	2	2
	Mounting Fixture not prope	0	0	0	0	0	0	0	0	0	0	0	0	0
	RJ45 Connector Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Print Not coming	0	0	0	0	0	0	0	0	0	0	0	0	0
	PC Hang	0	0	0	0	0	0	0	0	0	0	0	0	0
	MOXA Card not operated	0	0	0	0	0	0	0	0	0	0	0	0	0



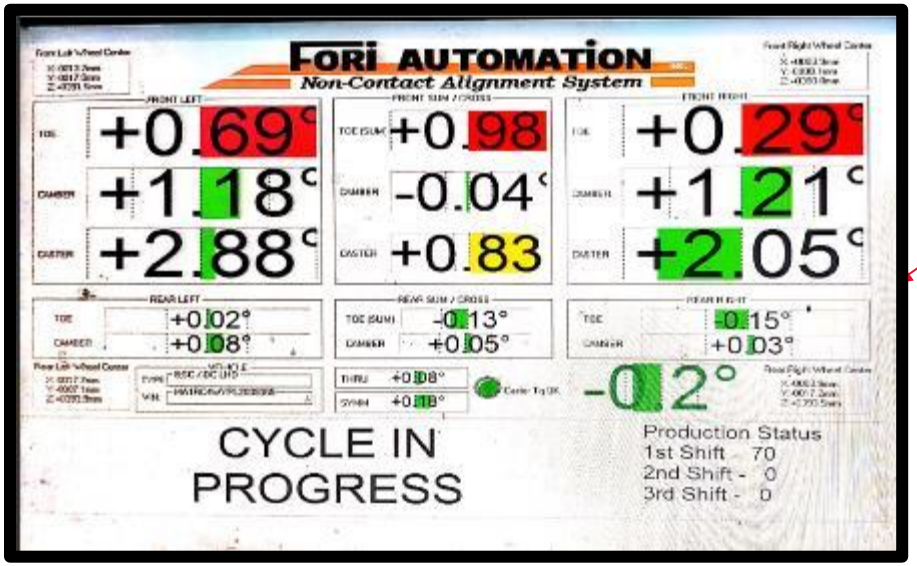
Caster Tool Readings Stuck

Last 1 Year Data analyze and Categories in different causes

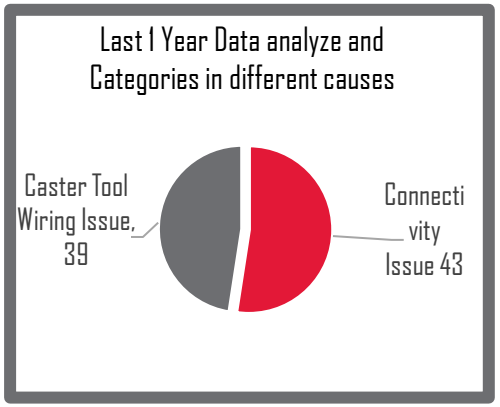
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Communication (Network) issue	Visual check for readings on machine screen	Data analysis done for last 3 Year – 43% of the problem is due to connectivity issue	**Hypothesis valid**



Camber and Toe readings come from Camera through network cable, so readings gets stuck in case of communication failure



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
6	PC hang	Visual checking with OK ,NOT OK	Data analysis done for last year – found Ok	**Hypothesis invalid**

Wheel Alignment Machine

Problem Monitoring Chart

Equipment name	Problem/Concern	18-Apr	18-May	18-Jun	18-Jul	18-Aug	18-Sep	18-Oct	18-Nov	18-Dec	18-Jan	18-Feb	18-Mar	18-Apr
Wheel Alignment Machine	New Operator	0	0	0	0	0	0	0	0	0	0	0	0	0
	Untrained Operator	0	0	0	0	0	0	0	0	0	0	0	0	0
	SOS not followed	0	0	0	0	0	0	0	0	0	0	0	0	0
	SOS inadequate	0	0	0	0	0	0	0	0	0	0	0	0	0
	Cradle not working	0	0	0	0	0	0	0	0	0	0	0	0	0
	Input voltage Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Display Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Program Wrong	0	0	0	0	0	0	0	0	0	0	0	0	0
	Barcode not updated	0	0	0	0	0	0	0	0	0	0	0	0	0
	Printer Cable Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDMI Cable Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Connectivity(Network) Issue	2	3	4	2	3	5	4	2	1	3	4	5	2
	Mounting fixture not	0	0	0	0	0	0	0	0	0	0	0	0	0
	Scanner not working	0	0	0	0	0	0	0	0	0	0	0	0	0
	Caster Tool Wiring Issue	1	2	1	2	1	1	1	2	2	1	1	2	2
	Mounting Fixture not prope	0	0	0	0	0	0	0	0	0	0	0	0	0
	RJ45 Connector Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Print Not coming	0	0	0	0	0	0	0	0	0	0	0	0	0
	PC Hang	0	0	0	0	0	0	0	0	0	0	0	0	0
	MOXA Card not operated	0	0	0	0	0	0	0	0	0	0	0	0	0

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	Print not coming	Visual checking with OK ,NOT OK	Data analysis done for last 1 year – No such issue observed.	**Hypothesis Invalid**

Wheel Alignment Machine

Problem Monitoring Chart

Equipment name	Problem/Concern	18-Apr	18-May	18-Jun	18-Jul	18-Aug	18-Sep	18-Oct	18-Nov	18-Dec	18-Jan	18-Feb	18-Mar	18-Apr
Wheel Alignment Machine	New Operator	0	0	0	0	0	0	0	0	0	0	0	0	0
	Untrained Operator	0	0	0	0	0	0	0	0	0	0	0	0	0
	SOS not followed	0	0	0	0	0	0	0	0	0	0	0	0	0
	SOS inadequate	0	0	0	0	0	0	0	0	0	0	0	0	0
	Cradle not working	0	0	0	0	0	0	0	0	0	0	0	0	0
	Input voltage Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Display Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Program Wrong	0	0	0	0	0	0	0	0	0	0	0	0	0
	Barcode not updated	0	0	0	0	0	0	0	0	0	0	0	0	0
	Printer Cable Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDMI Cable Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Connectivity(Network) Issue	2	3	4	2	3	5	4	2	1	3	4	5	2
	Mounting fixture not	0	0	0	0	0	0	0	0	0	0	0	0	0
	Scanner not working	0	0	0	0	0	0	0	0	0	0	0	0	0
	Caster Tool Wiring Issue	1	2	1	2	1	1	1	2	2	1	1	2	2
	Mounting Fixture not prope	0	0	0	0	0	0	0	0	0	0	0	0	0
	RJ45 Connector Issue	0	0	0	0	0	0	0	0	0	0	0	0	0
	Print Not coming	0	0	0	0	0	0	0	0	0	0	0	0	0
	PC Hang	0	0	0	0	0	0	0	0	0	0	0	0	0
	MOXA Card not operated	0	0	0	0	0	0	0	0	0	0	0	0	0

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Connectivity (Network) Issue	Observation noted for every month 43% issues due to Connectivity issues	**Hypothesis valid**
2	Caster Tool Issue	Data analysis done for last 3 Year – 39% issue observed.	**Hypothesis valid**

2 Valid Cause

Analysis

5.2 Confirm the probable cause:

COMPARISON STUDY for Wheel Alignment Machine working and non working condition

Wheel Alignment Machine not working	Wheel Alignment Machine working

CAD LEVEL COMPARISON

- Cycle Start button pressed
- Vehicle present Associate work as per instruction
- Caster, Camber, Toe readings gets stuck

- Cycle Start button pressed
- Vehicle present.
- Associate work as per instruction
- Readings getting adjusted correctly

CONCLUSION :- **Every component functioning but reading data not coming due to connectivity issues.**

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Connectivity (Network) Issue	Data not getting transmitted from amplifier panel to machine panel	Amplifier panel not receiving data from digital cameras	Network Cable not able to transmit data during high production setup	Replace the network cable
2	Caster Tool Issue	Caster tool not giving data to machine	Data not getting transmitted to caster tool sub panel	Wiring gets beak down at round connector junction	1) Modify the current facility 2) Put stand by cable with connector assembly

6. Implement the corrective measures

Possible Remedies Identified

Three possible solutions identified to Wheel Alignment Machine not working issue.

Legends

High ↑ Low ↓

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	**1) Connectivity (Network) Issue** **2) Caster Tool Issue**	Modify the machine layout	High ↑	High ↑	No	High ↑	**Rejected**
		Put stand by cable with connector assembly for Caster Tool connection	Low ↓	High ↑	No	Low ↓	**Accepted**
		Replace the network cable	Low ↓	High ↑	No	Low ↓	**Accepted**

6. Implement the corrective measures

Possible Remedies Identified

SOLUTION PROPOSAL 1

1. Modify the machine layout.

Changing the machine layout and changing complete wiring design leads to high cost and involves complexity

CONCLUSION :- This solution involve high cost and then also total problem will not be eliminated.

PROPOSAL REJECTED

6. Implement the corrective measures

Possible Remedies Identified

SOLUTION PROPOSAL 2

2. Replace the network cable with latest CAT-7 B high speed transfer cable.

Replacing the existing network cable from camera to machine with high speed CAT-7 B type cable for quick transfer of data, so that path does not gets blocked in case of high production setup.

CONCLUSION :- This solution feasibility need to check from implementation point of view. This will eliminate Both the problems.

PROPOSAL ACCEPTED

6. Implement the corrective measures

Possible Remedies Identified

SOLUTION PROPOSAL 3

3. Stand by cable with connector assembly for Caster Tool connection

- Standby cable with connector assembly to be made and kept near the caster tool cable for instant switchover in case of prior cable failure.

CONCLUSION :- This solution feasibility need to check from implementation point of view. This will eliminate Both the problems.

PROPOSAL ACCEPTED

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Cable connection gets open during prolonged use and repair of cable takes time	Provide an additional cable which can be used instantly in case of cable failure	29.09.2019	Prateek Srivastava	CLOSED

Before

Additional cable with connector Assembly

After

7. Check & confirm the results & effects

7.1 Verify Results:

INFERENCE :- **As per data collected ,after action implementation no defect reported in last 5 months. Hence concern is eliminated after taking PCA action through MYB project.**

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	Taking suggestions for high speed cable	16.06.2019	Prateek Srivastava	CLOSED
2	Quotation taken of new cable	27.06.2019	Prateek Srivastava	CLOSED
3	Procurement of New cable	21.07.2019	Hari Nagar	CLOSED
4	Installation Of new cable	28.08.2019	Prateek Srivastava	CLOSED
5	PM Sheets	20.10.2019	Prateek Srivastava	CLOSED
6	Daily Checksheet	20.10.2019	Prateek Srivastava	CLOSED
7	MP Sheet	20.10.2019	Hari Nagar	CLOSED
8	CLITA Sheet	20.10.2019	Prateek Srivastava	CLOSED

Check & confirm the results & effects (Cont.....)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) 192 UPS production achieved
2) Minor Stoppages of Wheel Alignment machine eliminated.
2) STR loss due to Wheel Alignment machine failure is eliminated completely.

In tangible Benefits –

1) Associate moral improved
2) Get direct benefits in RFD
3) Learn the Gemba culture

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	2.0
Total Required Rework time per day	Min	20
Effective working time	Min	465
Operator required /day	nos	0.04
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	84.90
Avg no. of working days	days	25
Operator cost /month	Rs	2122.6
Operator Cost/year	Rs	25471.0

Check & confirm the results & effects (Cont.....)

7.4 : Cost Approval Mail

Sun 22-12-2019 15:47
DHANOKAR RAJESH
RE: Cost calculation of worker
To SRIVASTAVA PRATEEK - NASHIK PLANT I

Repair /Restoration Cost estimation		
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Operator cost /month	Rs	2122.6
Operator Cost/year	Rs	25471.0

From: SRIVASTAVA PRATEEK - NASHIK PLANT I
Sent: Sunday, December 22, 2019 2:38 PM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Subject: Cost calculation of worker

Dear Sir,

Please share the rework cost.

8. Standardization

8.1 Updation of documents: e.g. MP Sheet, PM Sheets, Daily check sheet.

SN	Document	Actual date	Responsibility	Status
1	MP Sheets	20.10.2019	Hari Nagar	CLOSED
2	PM Sheets	21.10.2019	Prateek Srivastava	CLOSED
3	Daily check sheet	20.10.2019	Prateek Srivastava	CLOSED
4	CLITA Sheet updated	20.10.2019	Prateek Srivastava	CLOSED

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Training given to Maintenance Associate.	20.10.2019	Prateek Srivastava	Closed

8.3 Additional Audit & check

1	Check point added in DWM	09.09.19	Prateek Srivastava	Closed

8. Standardization

8.1 Updation of documents: e.g. MP Sheet, PM Sheets, History card.

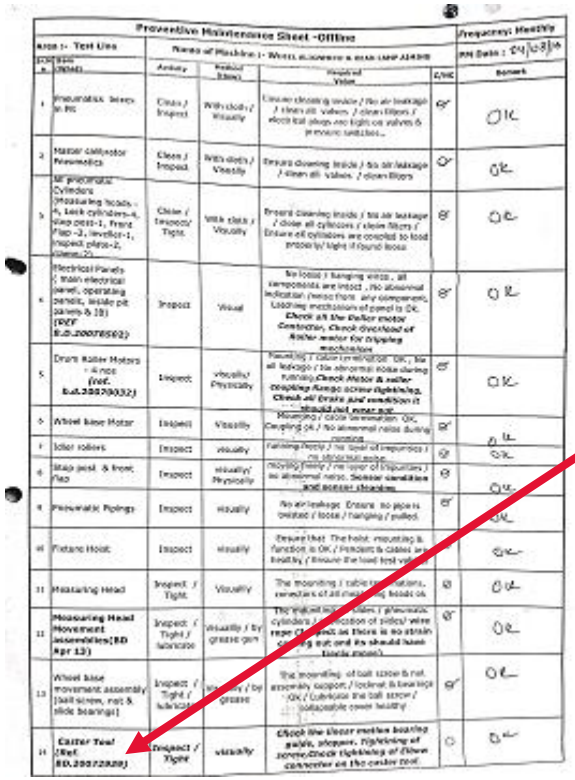
SN	Document	Actual date	Responsibility	Status
1	MP Sheet	20.10.2019	Hari Nagar	CLOSED

Mahindra Rise. Automotive Sector Nasik-422007	Maintenance Prevention (MP) Sheet	PU/Dept : SXTCF Cell : CAI Line
PU	Scorpio TCF	
Cell	CAI	
Machine	Wheel Alignment Machine	
Machine No.		
Phenomenon Observed	Readings Stuck	
Date	20/10/2019	
Why-Why Analysis	Data not getting transmitted from amplifier panel to machine panel	
	⬇	
	Amplifier panel not receiving data from digital cameras	
	⬇	
	Network Cable not able to transmit data during high production setup	
Root Cause	Cable not able to transfer data simultaneously during high setup	
Immediate action taken to start machine/equipment	Replace the network cable with latest CAT-7 B high speed transfer cable.	

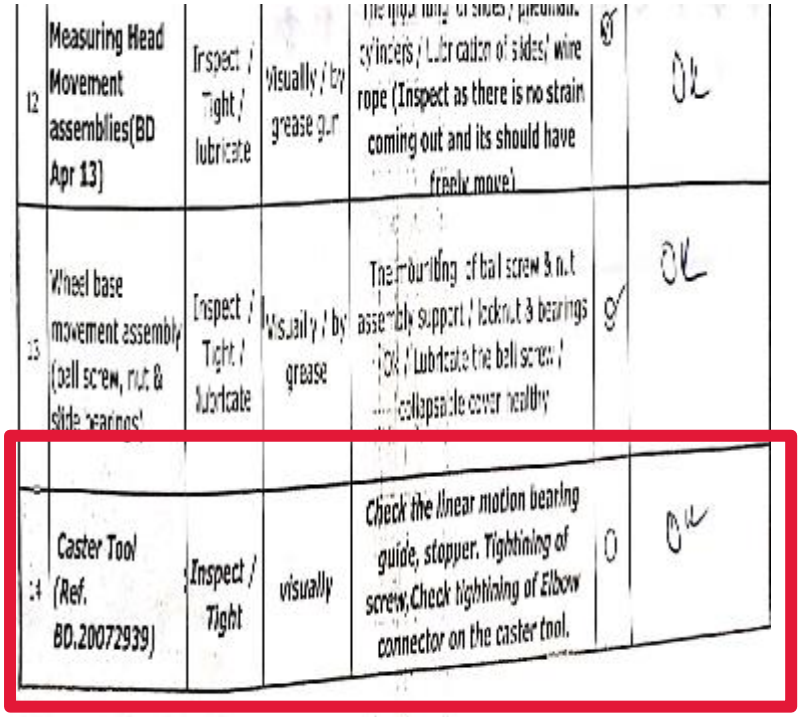
8. Standardization

8.1 Updation of documents: e.g. MP Sheet, PM Sheets, History card.

SN	Document	Actual date	Responsibility	Status
2	PM Check Sheet Updated	21.10.2019	Prateek Srivastava	CLOSED



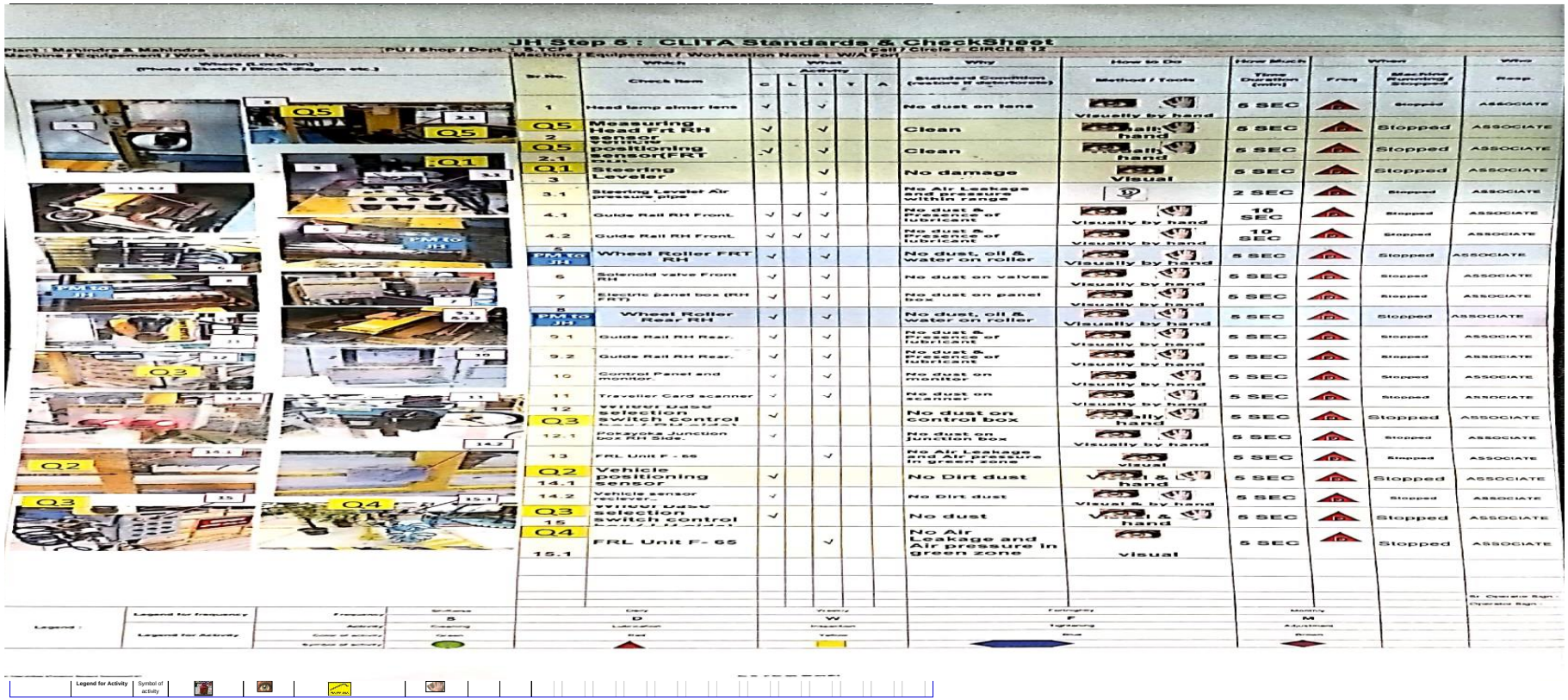
Point Added in Check sheet



8. Standardization

8.1 Updation of documents: e.g. MP Sheet, PM Sheets, History card.

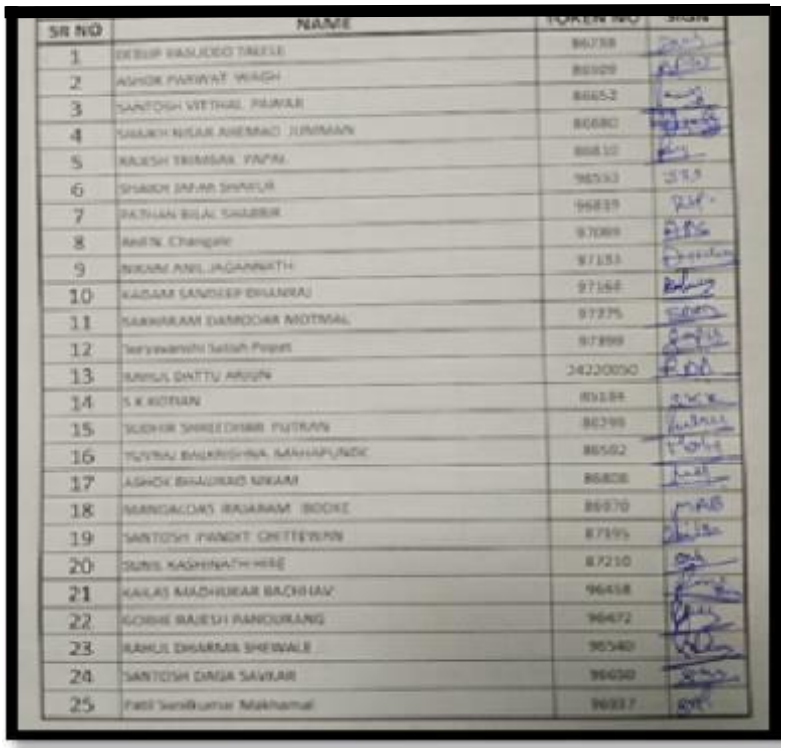
SN	Document	Actual date	Responsibility	Status
4	CLITA Sheet updated	20.10.2019	PRATEEK SRIVASTAVA	CLOSED



8. Standardization

8.2 Training:

Action Plan	Target Dt	Resp.	Status
Training given to Maintenance Associate.	20.10.2019	Prateek Srivastava	Closed



SR NO	NAME	TOKEN NO	SIGN
1	DEEPAK MASUDDO TAKTE	86738	
2	ASHOK PARWAT WAGH	86909	
3	SANTOSH VITHAL PAWAR	86652	
4	SHAIKH NISAR AHEMAD JUMMAN	86680	
5	RAJESH TRIMBAK YAPAN	86810	
6	SHAIKH JAFAR SHAFIUR	96533	
7	PATHAN BILAL SHABBIR	96839	
8	RAFIK Changale	97089	
9	NIKAM ANIL JAGANNATH	97153	
10	KADAM SANDEEP BHAURAO	97168	
11	RAMNIKLAL DAMODAR MOTIWAL	97375	
12	Yerawanshi Satish Popat	97899	
13	RAHUL DATTU ARGUN	24220050	
14	S K KOTIAN	85184	
15	SUDHIR SHREEDHAR PUTRAN	86299	
16	PUNTAV BALKRISHNA MAHAPUNDIK	86502	
17	ASHOK BHAUSAHEB NIKAM	86808	
18	MANGALDAS IRAJANAM BODKE	86970	
19	SANTOSH PANDIT CHITTEWAN	87195	
20	SUNIL KASHINATH HIRE	87210	
21	KAILAS MADHUKAR BACHHAV	96458	
22	GOBHE RAJESH PANDURANG	96472	
23	RAHUL DHARMA SHEWALE	96540	
24	SANTOSH DAGA SAVLAR	96650	
25	Patil Sunilkumar Makhamal	96837	

9. Summarize & Plan future actions

9.1 Conclusion:

With implementation of additional cable and replacement of existing cable with high speed cable WA stage , WA not working concern is eliminated.

9.2 Horizontal Deployment:

Input given to BTCF.

9.3 Lessons Learnt:

Things Gone Wrong : Initially at the time of Installation failure due to due Design standardization of machine were not consider.

Things Gone Right :

Considering the role of data transfer speed in wheel alignment machine helped us in getting results

Thank you

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